

Worksheet 7C: Eigenvalues and Eigenvectors

Series 7: Linear Algebra
AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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What This Worksheet Is About

Most vectors, when multiplied by a matrix, change both length and direction. But some special vectors — eigenvectors — only get scaled. These are the “natural axes” of a matrix. The stationary distribution of a Markov chain is an eigenvector; Principal Component Analysis is pure eigenvalue decomposition; Google’s PageRank is an eigenvector computation. This worksheet builds the concept geometrically, then develops the full algebra.

Part A: Geometric Intuition (8 problems)

If $A\mathbf{v} = \lambda\mathbf{v}$ for a non-zero vector $\mathbf{v}$, then $\mathbf{v}$ is an **eigenvector** with **eigenvalue** λ .

1. $A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$. (a) Compute $A \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $A \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. (b) Are these eigenvectors? (c) Compute $A \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ — is it an eigenvector? (d) What do the eigenvalues tell you about what A does?
2. $A\mathbf{v} = 5\mathbf{v}$ for every $\mathbf{v}$. What matrix is A ? How many eigenvectors does it have?
3. Rotation matrix $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (90° rotation). (a) Can any real vector stay pointing the same way after a 90° rotation? (b) What does this say about real eigenvalues of R ?
4. Reflection matrix $F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. (a) Find two eigenvectors by inspection. (b) What are their eigenvalues? (c) Interpret the negative eigenvalue geometrically.
5. Show that if $\mathbf{v}$ is an eigenvector of A with eigenvalue λ , then $2\mathbf{v}$ is also an eigenvector with eigenvalue λ .
6. Show that if $\mathbf{v}$ and $\mathbf{w}$ are both eigenvectors with the *same* eigenvalue λ , then $\mathbf{v} + \mathbf{w}$ is also an eigenvector (if non-zero).
7. If $\mathbf{v}$ is an eigenvector of A with eigenvalue $\lambda \neq 0$, what is the corresponding eigenvalue of A^{-1} ?
8. I has every vector as an eigenvector. O also has every vector as an eigenvector. What are their eigenvalues?

Part B: Finding Eigenvalues (10 problems)

The eigenvalues satisfy $\det(A - \lambda I) = 0$ (the **characteristic equation**).

For 2×2 : $\det(A - \lambda I) = \lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$.

Find characteristic polynomial and eigenvalues:

9. $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

10. $B = \begin{bmatrix} 5 & -1 \\ 2 & 3 \end{bmatrix}$

11. $C = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$

12. $D = \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}$ (repeated eigenvalue)

13. $E = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (complex eigenvalues $\pm i$)

14. For problems 9 and 11, verify: $\lambda_1 + \lambda_2 = \text{tr}(A)$ and $\lambda_1 \lambda_2 = \det(A)$.

15. If $\det(A) = 0$, what must be true about one eigenvalue? What does this say about invertibility?

16. Find eigenvalues of the triangular matrix $T = \begin{bmatrix} 2 & 4 & -1 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{bmatrix}$. (The eigenvalues of a triangular matrix are its diagonal entries.)

17. An $n \times n$ matrix has exactly n eigenvalues (counting multiplicity, including complex). What is the maximum number for a 3×3 matrix?

18. Is $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ positive definite (all eigenvalues positive)?

Part C: Finding Eigenvectors (10 problems)

For each eigenvalue λ , solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ via row reduction.

Find eigenvalues and eigenvectors:

19. $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

20. $B = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$

21. $C = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$ (diagonal — what do you expect?)

22. $D = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ (repeated $\lambda = 2$; only one independent eigenvector)

23. Verify your eigenvectors for problem 19 by substituting into $A\mathbf{v} = \lambda\mathbf{v}$.
24. $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. If $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector, find λ . Find the other eigenvalue and eigenvector.
25. A symmetric matrix always has real eigenvalues and orthogonal eigenvectors for distinct eigenvalues. Verify orthogonality for problem 24.
26. For the Dublin weather matrix $P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$: (a) Find eigenvalues. (b) Find eigenvector for $\lambda = 1$. (c) Compare to the stationary distribution from ws07d.
27. Covariance matrix $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$: find eigenvalues and eigenvectors. Which eigenvector is the first principal component?
28. What fraction of total variance is captured by the first principal component in problem 27?

Part D: Diagonalisation (8 problems)

If A has n linearly independent eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$:

$$A = P\Lambda P^{-1}$$

where $P = [\mathbf{v}_1 \mid \dots \mid \mathbf{v}_n]$ and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. This gives $A^k = P\Lambda^k P^{-1}$.

29. For $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, form P and Λ . Verify $A = P\Lambda P^{-1}$.
30. Use diagonalisation to compute A^3 (problem 19's matrix).
31. Diagonalise $P_{\text{chain}} = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$. Compute P_{chain}^n as $n \rightarrow \infty$. What does each row converge to?
32. Why can't $D = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ be diagonalised?
33. The Spectral Theorem: every symmetric matrix $A = Q\Lambda Q^T$ with Q orthogonal. Why is the covariance matrix always symmetric? What does this guarantee about PCA?
34. If A has eigenvalues λ_1, λ_2 , find the eigenvalues of: (a) $3A$ (b) $A + 5I$ (c) A^2 (d) A^{-1}
35. A matrix has eigenvalues $2, -1, 3$. Find: (a) trace (b) determinant (c) Is it invertible?
36. **PCA and dimensionality reduction.** A covariance matrix has 1000 eigenvalues; the top 10 capture 95% of total variance. (a) Why is this useful for ML? (b) What is lost? (c) How does this relate to the rank concept from ws07b?

Part E: Connections (4 problems)

37. Show that the Markov steady-state equation $\boldsymbol{\pi}P = \boldsymbol{\pi}$ is equivalent to finding a left eigenvector of P with $\lambda = 1$ (or equivalently a right eigenvector of P^T).
38. **PageRank.** The ranking vector is the eigenvector of the modified link matrix with $\lambda = 1$. Why must this eigenvector be unique for a reliable search ranking?
39. For $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$, find $A^T A$. The square roots of the eigenvalues of $A^T A$ are the singular values of A .
40. Reflection: (a) ws07a: what does A do? ws07b: how to solve $A\mathbf{x} = \mathbf{b}$? ws07c: what are A 's natural axes? How do these three questions fit together? (b) The eigenvector with the largest eigenvalue is the direction of maximum stretch. Why does this make it the first principal component?